

MadhuTwin: a hybrid digital twin for type 2 diabetes

Technical report · Happiest Health Digital Twin Challenge 2026, Phase 1 · Team: TODO(team)

Abstract. MadhuTwin is a hybrid digital twin for people with type 2 diabetes. A mechanistic glucose–insulin model is personalised to each person from their EHR and first days of data, kept synchronised with their continuous glucose monitor by an Unscented Kalman Filter, and run forward to forecast glucose and simulate what-ifs. A physics-gated neural network (TwinNet) fuses the EHR, wearable streams and the twin's own forecast; it is fine-tuned on each person and averaged with a gradient-boosted model. On 199 unseen synthetic Indian patients, the 60-minute forecast error is 18.0 mg/dL against 35.8 for persistence, 99.5% of forecasts are clinically acceptable (Clarke A+B), and 90% of sustained spikes above 180 mg/dL are flagged a median 110 minutes ahead with at most one false alert per patient-day. On real people with patient-level cross-validation, CGMacros (45): 22.4 vs 28.3 mg/dL; BIG IDEAs (16): 19.0 vs 23.0 mg/dL; ShanghaiT2DM (109): 21.1 vs 27.5 mg/dL (MadhuTwin vs persistence, 60 min). With a week of personal data, the 2-hour error falls from 31.4 to 27.2 mg/dL. The synchronised twin detects illness days from CGM alone (AUROC 0.89). Results are reported with patient-bootstrap confidence intervals, calibration and decision curves, a subgroup audit and stated limitations.

1. Problem and use case

India has about 101 million adults with diabetes and 136 million with prediabetes (ICMR-INDIAB, 2023). Care is visit-based: a quarterly HbA1c summarises the past, while the dangerous moments (post-meal spikes, night-time lows on sulfonylureas or premixed insulin, loss of control during illness) happen between visits. CGMs and smartwatches now record them, but a care team cannot watch hundreds of streams. MadhuTwin targets **remote monitoring by a diabetes care team**: forecast glucose 5–120 minutes ahead with an 80% interval, estimate the probability of a sustained excursion above 180 or below 70 mg/dL within two hours, rank the panel by risk, explain each alert, and let the clinician test meals, walks, sleep or illness on the patient's own twin.

2. System overview

MadhuTwin architecture: two data streams fused into a living digital twin

Type 2 Diabetes · predicts glucose spikes and hypoglycaemia up to 2 hours ahead · doctor-facing dashboard

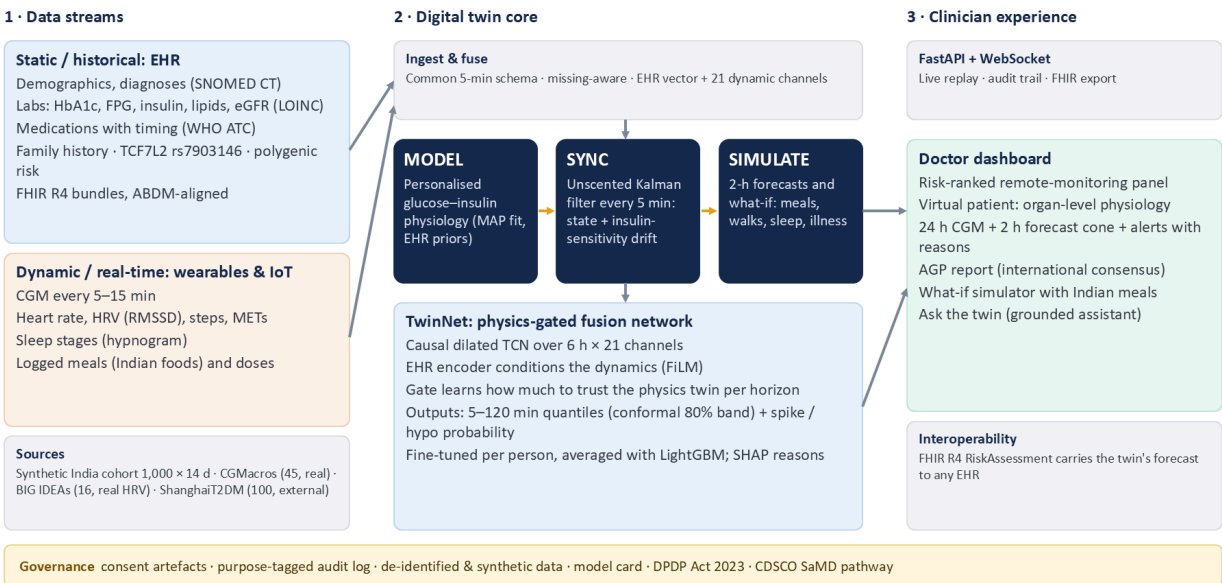


Figure 1. Two data streams (EHR and wearables) are ingested into one 5-minute schema; the twin models, synchronises and simulates; TwinNet fuses everything; the dashboard and a FHIR R4 RiskAssessment deliver the result.

The system has three digital-twin properties. **Model**: a personalised mechanistic model of glucose–insulin physiology. **Sync**: the model's hidden state and a drifting insulin-sensitivity factor are re-estimated from every CGM reading. **Simulate**: the synchronised twin is run forward for forecasts and counterfactual what-ifs. On top, TwinNet learns the residual the physics misses and how much to trust the physics at each horizon.

3. Data

Source	People	Streams	Role
Synthetic India-calibrated cohort (this project)	1,000 × 14 days	FHIR R4 EHR (labs, ATC drugs, SNOMED diagnoses, genetics), CGM, HR, HRV, steps, sleep, meals, doses	Training at scale; controlled ablations; illness ground truth
CGMacros (PhysioNet, CC BY-NC-SA 4.0)	45	Dexcom/Libre CGM, Fitbit HR and steps, meal macros, labs	Real-world validation; twin-validity analysis
BIG IDEAs (PhysioNet, ODC-By 1.0)	16	Dexcom G6 CGM, Empatica E4 heart rate and beat intervals (HRV computed), food logs	Real wristband HR and HRV
ShanghaiT2DM (Figshare, CC BY 4.0)	100 (109 recordings)	15-min CGM, diet, insulin and oral drugs, labs; no wearables	External population; missing-modality test

The synthetic EHR follows Indian epidemiology (status mix from ICMR-INDIAB, South-Asian BMI phenotype, Indian prescribing such as premixed 30/70 insulin, gliclazide and teneligliptin) and regional meals from the Indian Food Composition Tables. Crucially, the wearable streams are generated by the same simulated body, and labs are measured from it (HbA1c from mean glucose via ADAG), so the two streams are causally coupled. Realistic imperfections are injected: unlogged meals, carbohydrate misestimation, CGM noise and compression lows, missed doses, illness and stress days. Physiological parameter distributions come from twins fitted to the 45 CGMacros participants.

Synthetic CGM statistics vs real people (5th-95th, IQR, median); the Indian T2D cohort is deliberately sicker (median HbA1c 8.0% vs 7.1%)

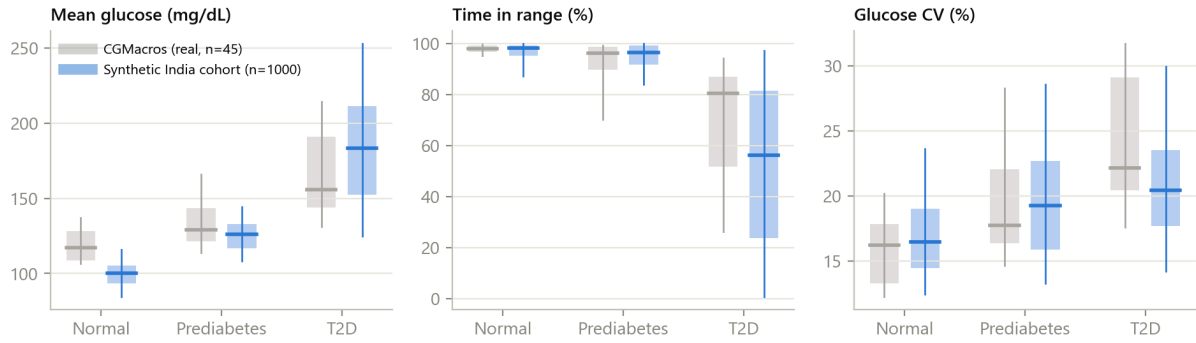


Figure 2. Synthetic CGM statistics against real people by glycaemic status.

4. Methods

4.1 Mechanistic model

$$\begin{aligned}
 dG/dt &= -SG (G - G_b \cdot \text{egp}) - SI \cdot m_{si} \cdot X \cdot G + Ra(t) - k_{ex} \cdot E1 \cdot G - \text{renal}(G) + \text{counterreg}(G) \\
 dX/dt &= -p2 (X - (I - I_b)) \\
 dI/dt &= -n (I - I_b) + \beta \cdot m_{\beta} \cdot [G - G_b]_+ + k_{inc} \cdot m_{inc} \cdot Ra(t) + I_{ex}(t) + SU(t) \\
 dGi/dt &= (G - Gi) / \tau_{isf} \quad (\text{interstitial glucose seen by the CGM}) \\
 dE1/dt &= (A(t) - E1)/10 ; \quad dE2/dt = (A(t) - E2)/360 \quad (\text{acute and lingering exercise})
 \end{aligned}$$

Meal appearance $Ra(t)$ depends on carbohydrate, glycaemic index, fat and fibre. Time-varying multipliers encode sleep loss, illness and stress (m_{si}), the dawn phenomenon (egp), sulfonylurea secretion, DPP-4 incretin enhancement (m_{inc}), the SGLT2-lowered renal threshold, and rapid, regular, basal and premixed insulin kernels. A Numba-compiled integrator (verified against the NumPy reference) makes a patient fit about 12× faster.

4.2 Personalisation (Model)

Five parameters (fasting set-point G_b , insulin sensitivity SI , beta-cell response β , dawn amplitude, absorption-speed scale) are fitted by maximum a posteriori estimation with EHR-derived priors, using a multiple-shooting objective over 4-hour windows (1-hour burn-in, 3 hours scored). On the 45 CGMacros participants the twin's error falls from 37.7 mg/dL with EHR priors to 26.6 mg/dL once personalised, and the fitted SI and β track laboratory HOMA-IR and HbA1c (Figure 3).

Twins fitted to 45 real people recover known physiology from CGM

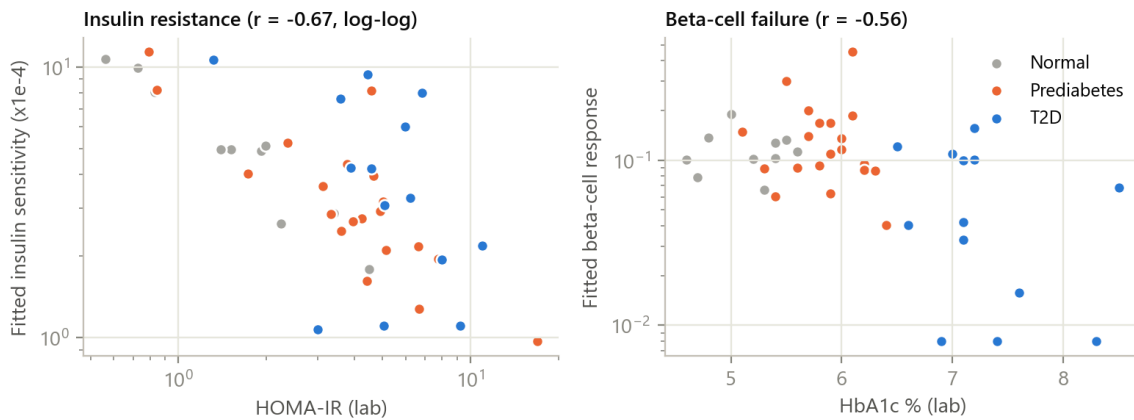


Figure 3. Fitted parameters recover known physiology on 45 real people.

4.3 Synchronisation (Sync)

An Unscented Kalman Filter estimates seven states every 5 minutes: the six model states and a log insulin-sensitivity multiplier that follows a bounded random walk. A sustained fall in that multiplier is surfaced as an insight (possible infection, stress or steroid use).

4.4 TwinNet and the MadhuTwin ensemble (Simulate)

TwinNet reads 6 hours $\times$ 21 dynamic channels (CGM, heart rate, HRV, steps, METs, sleep, carbohydrate and insulin on board, logged events, time of day) through a causal dilated temporal convolution (dilations 1–16). An EHR encoder conditions the sequence features through FiLM, so the same meal is read differently for different bodies. The synchronised twin's 2-hour forecast and hidden state enter through a learned per-horizon gate; heads output P10/P50/P90 glucose at 5–120 minutes and the probabilities of a spike or a low within 2 hours. Modality dropout lets one model run without wearables or EHR, and conformalised quantile regression calibrates the 80% band. The learned trust in physics rises from 0.71 at 30 minutes to 0.92 at 2 hours.

The twin then learns the individual: TwinNet is fine-tuned for a few epochs on the person's own earlier windows only, and its median and event probabilities are averaged with a LightGBM model on hand-crafted features of the same inputs. LightGBM's TreeSHAP values give the plain-language reasons attached to every alert. Both models are trained with class weighting so that rare events are learned, which inflates raw probabilities; the ensemble's risks are therefore recalibrated with Platt scaling, fitted on validation patients for the synthetic cohort and, for each real-data site, on other patients only (cross-fitted by fold in evaluation).

5. Evaluation protocol

- **Splits.** Patient-level throughout. Synthetic: 70/10/20 split by patient; scored only on days 8–14, after the twin and TwinNet were personalised on days 1–7. Real data: 5-fold cross-validation grouped by patient; models are pretrained on synthetic data and fine-tuned on training folds only (sim-to-real).
- **Forecasts.** RMSE, MARD, Clarke Error Grid zones and 80% interval coverage at 30, 60, 90 and 120 minutes; confidence intervals by patient bootstrap.
- **Events.** A spike is glucose above 180 mg/dL (a low: below 70) sustained for at least 15 minutes within the next 2 hours. AUROC and AUPRC; the share of excursions alerted in the 2 hours before onset at the operating point with at most one false-alert episode per patient-day, and the median lead time. Alert thresholds on real data are chosen on the training folds.
- **Clinical utility.** Calibration curves, decision-curve net benefit (Vickers and Elkin, 2006) and a subgroup audit.

6. Results

6.1 Forecasting on unseen synthetic patients

Method	RMSE 30	RMSE 60	RMSE 120	MARD 60 %	Clarke A+B 60 %	PI80 cov. 60 %
Persistence	22.9	35.8	50.4	14.7	98.2	–
TwinNet hybrid	13.6	19.0	24.9	8.2	99.4	–
LightGBM fusion	13.1	18.4	24.4	8.3	99.5	–
TwinNet personalised	13.4	18.6	24.2	8.1	99.5	79
MadhuTwin ensemble	13.0	18.0	23.3	8.0	99.5	79

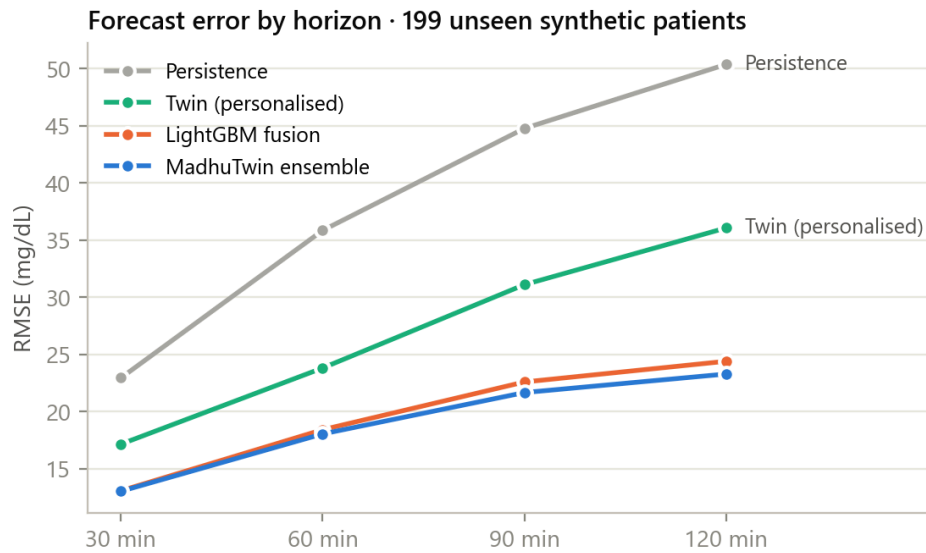


Figure 4. Forecast error by horizon on unseen synthetic patients.

6.2 Adverse-event prediction

Method	AUROC	AUPRC	Prev. %	Caught at ≤ 1 FA/day %	Lead (min)
TwinNet hybrid	0.956	0.859	20.0	90	110
LightGBM fusion	0.956	0.860	20.0	90	110
TwinNet personalised	0.947	0.829	20.0	89	110
MadhuTwin ensemble	0.961	0.871	20.0	90	110

Spike above 180 mg/dL within 2 hours. Hypoglycaemia: AUROC 0.968 for MadhuTwin ensemble.

6.3 Does fusing the two streams help?

Streams (LightGBM retrained)	RMSE 60	Spike AUROC
CGM only	24.87	0.923
CGM + EHR	23.82	0.931
CGM + wearables	23.99	0.930
CGM + meals/meds	21.38	0.937
CGM + wearables + meals/meds	21.12	0.941
Both streams (dynamic + EHR)	20.12	0.949
Both streams + physics twin	18.44	0.955

Every stream adds information, and the EHR helps most in combination: it tells the model whose body the wearables describe. Adding the physics twin's forecast as a feature cuts the 60-minute error from 24.9 (CGM only) to 18.4 mg/dL.

6.4 The twin learns you

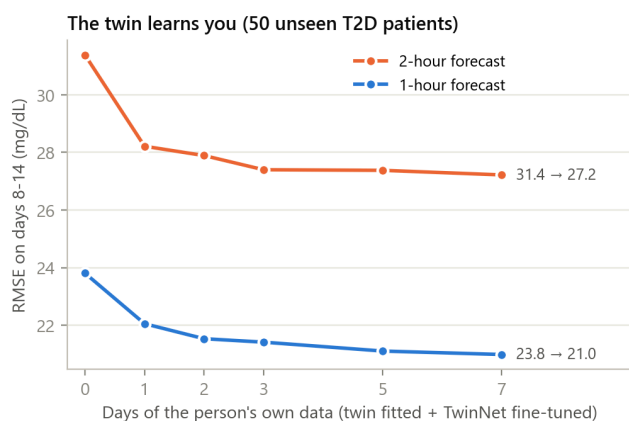


Figure 5. Error on days 8–14 against days of personal data used to fit the twin and fine-tune TwinNet (50 unseen T2D patients; day 0 = population model).

Days of personal data	RMSE 60	RMSE 120
0	23.81	31.38
1	22.05	28.21
2	21.53	27.89
3	21.41	27.40
5	21.10	27.38
7	20.98	27.22

6.5 Calibration, net benefit and subgroups

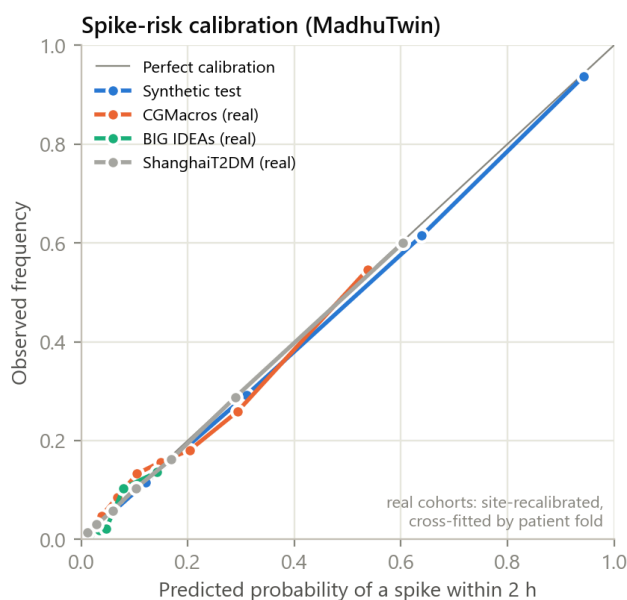


Figure 6. Calibration of spike risk (deciles of predicted risk).

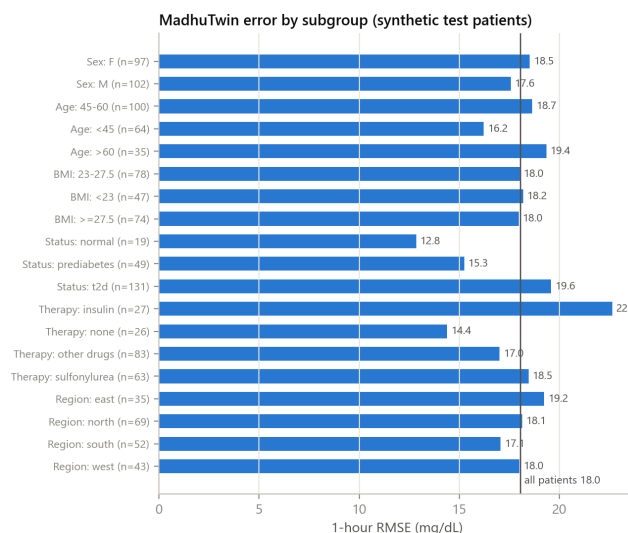


Figure 7. One-hour error by subgroup (synthetic test patients; groups with at least 5 patients).

Decision curves: spike alerts give net benefit across thresholds

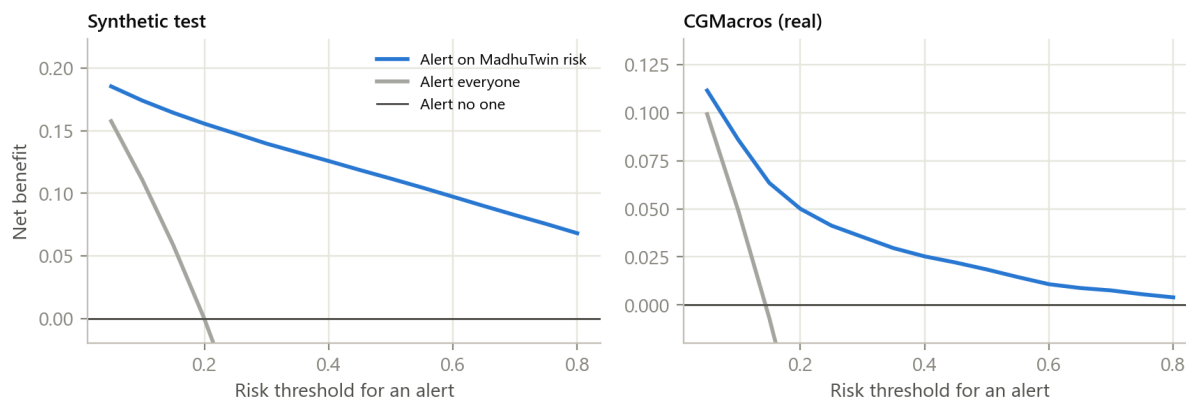


Figure 8. Decision curves: net benefit of alerting on MadhuTwin's spike risk against alerting everyone or no one.

Across 19 subgroups the 1-hour error ranges from 12.8 to 22.7 mg/dL.

6.6 Real people and an external population

Cohort (recordings)	Persistence RMSE 60	MadhuTwin RMSE 30	RMSE 60	RMSE 120	Clarke A+B 60 %	Spike AUROC	Caught at ≤ 1 FA/day %
CGMacros (45)	28.3	15.9	22.4	27.6	99.5	0.822	82
BIG IDEAs (16)	23.0	13.9	19.0	21.8	99.3	0.765	58
ShanghaiT2DM (109)	27.5	12.2	21.1	30.1	98.9	0.893	89

All real-data results use 5-fold patient-level cross-validation. A model trained only on synthetic data does not transfer to a new population without a short fine-tune (zero-shot rows in the evaluation report), while synthetic pretraining followed by fine-tuning on a few real patients is the regime we recommend.

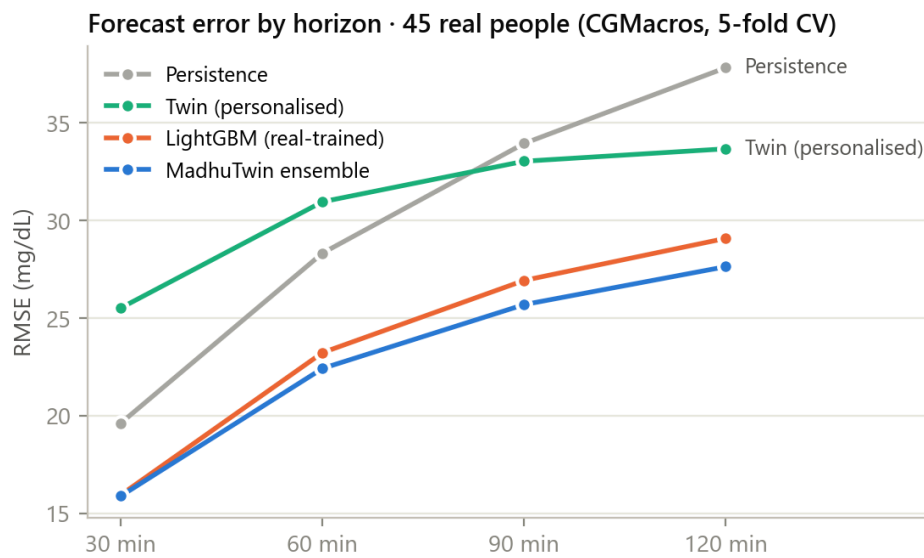


Figure 9. Forecast error by horizon on 45 real people (CGMacros, 5-fold patient CV).

6.7 Illness detection by the synchronised twin

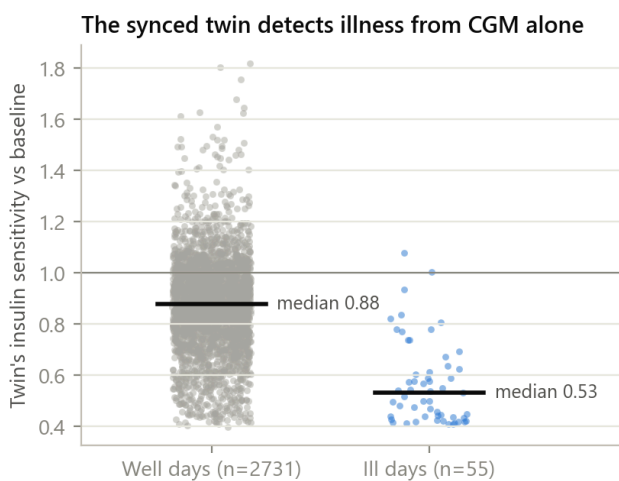


Figure 10. Daily insulin sensitivity estimated by the UKF on ill and well days.

Using the daily mean of the UKF's insulin-sensitivity multiplier as a score, ill days are separated from well days with AUROC **0.885** (55 ill days of 2786); the multiplier averages 0.57 on ill days against 0.87 on well days, without any symptom being entered.

6.8 CGM-light: an affordability mode

After one week of CGM, only 4 fingersticks a day and a smartwatch remain. The synchronised twin estimates continuous glucose with a MARD of 6.4% (carrying the last fingerstick forward: 23.8%) and weekly time in range within ± 3.8 percentage points, on 199 unseen synthetic patients. Because the simulator shares the twin's structure, this is an upper bound until tested on real CGM-off data.

6.9 Can the twin be trusted for a whole day?

The therapy simulator (section 7) runs the mechanistic twin open-loop for 24 hours, so we test exactly that: from the synced state at the start of every day after calibration, the twin replays the day with the logged meals, doses and activity, and the result is compared with the CGM.

Cohort	Days	Median MAE	Daily-mean error	TIR error (pp)	Good %	Poor %
Synthetic test	350	13.7	5.5	4.9	69	2
CGMacros	212	17.9	7.0	2.8	55	14
BIG IDEAs	55	14.4	5.1	0.7	80	2
ShanghaiT2DM	511	26.2	11.7	7.3	25	29

The twin tracks daily mean glucose and time in range closely but rarely reproduces short lows. The simulator therefore reports means, time in range and time above range, shows this replay check for each patient (with a warning when agreement is poor), and leaves hypoglycaemia warnings to the validated 2-hour forecast.

6.10 Robust to missing data

One deployed model (trained with modality dropout) scored with data removed at inference, 129,956 synthetic test forecasts:

Missing at inference	RMSE 60	RMSE 120	Spike AUROC
All data	19.04	24.95	0.956
No smartwatch (wearables missing)	19.51	26.05	0.950
No EHR	19.40	25.33	0.953
No meal or dose logs	19.09	25.13	0.955
No physics twin	21.21	27.41	0.946
Only CGM (no smartwatch, EHR, logs or twin)	27.25	31.97	0.916
10% of past CGM readings lost	19.08	24.98	0.955
25% of past CGM readings lost	19.12	25.08	0.954
50% of past CGM readings lost	19.29	25.44	0.951

6.11 Speed

On one laptop CPU (8 threads, no GPU): a 2-hour forecast takes 2.2 ms (9523 per second batched); personalising a new patient takes 1.8 s (twin fit on 7 days plus TwinNet fine-tuning); syncing the twin costs 11.0 ms per patient-day; and a 24-hour therapy simulation takes 0.20 ms per plan. Compute alone would refresh about 2.2 million patients every 5 minutes, so data handling, not the model, limits scale.

7. Dashboard, interoperability and governance

The doctor-facing dashboard (React, TypeScript, ECharts) shows a risk-ranked panel, a virtual-patient view (24-hour CGM, the 2-hour forecast cone, the physics twin's projection, alerts with reasons and each patient's own alert track record, organ-level physiology), an AGP report, a what-if simulator with Indian meals, a 24-hour therapy simulator (insulin and sulfonylurea doses and timing, DPP-4 and SGLT2 inhibitors, with stress tests such as a skipped lunch,

and a dose-response table), and a grounded "Ask the twin" assistant (offline by default). Forecasts are exported as FHIR R4 RiskAssessment resources with SNOMED CT, LOINC and WHO ATC codes, aligned with ABDM. The design follows the DPDP Act 2023: consent, purpose limitation, data minimisation and a purpose-tagged audit trail of every access. Only synthetic and openly licensed, de-identified data are used.

8. Limitations

- Synthetic patients come from a simulator with the same structure as the twin, which flatters physics-based methods there; real and external results are the primary evidence.
- The real cohorts are small (45, 16 and 100 people) and none is Indian. A prospective pilot in Indian clinics is required.
- Real-world hypoglycaemia events are rare and mostly in people without diabetes (CGMacros), so hypoglycaemia precision on real data is not established.
- Raw risks from class-weighted training are too high; each site must recalibrate on its own past patients before going live. On BIG IDEAs (people without diabetes, few spikes) spike prediction is weak (AUROC about 0.77) and adds little net benefit.
- Unlogged meals dominate 2-hour error; forecasts never use future information. Alert thresholds must be tuned with clinicians.
- The 24-hour therapy simulator rests on the mechanistic twin: it tracks daily mean glucose and time in range but rarely short lows, does not model metformin (which acts on the fasting set-point over weeks), and has not yet been validated against real dose changes.
- Subgroup performance is audited on synthetic patients only. This is research software, not a medical device; the regulatory path is CDSCO Software as a Medical Device with prospective validation.

References

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